Agile Development Process and Devops Lab

Assessment – 5

Spring Boot CI/CD Pipeline (GitHub + Jenkins + Docker + Kubernetes)

PREETHIKSHABALA V

23MIS0085

o (1).zip.369\... | HelloController.java | application.yaml | Dockerfile | dep

C: > Users > PREETHIKSHABALA V > Downloads > demo > src > main > java > com > example > demo

```java
package com.example.demo;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    @GetMapping("/")
    public String home() {
        return "App Running Successfully 🚀";
    }
}
```

o (1).zip.369\... | HelloController.java | application.yaml | Dockerfile

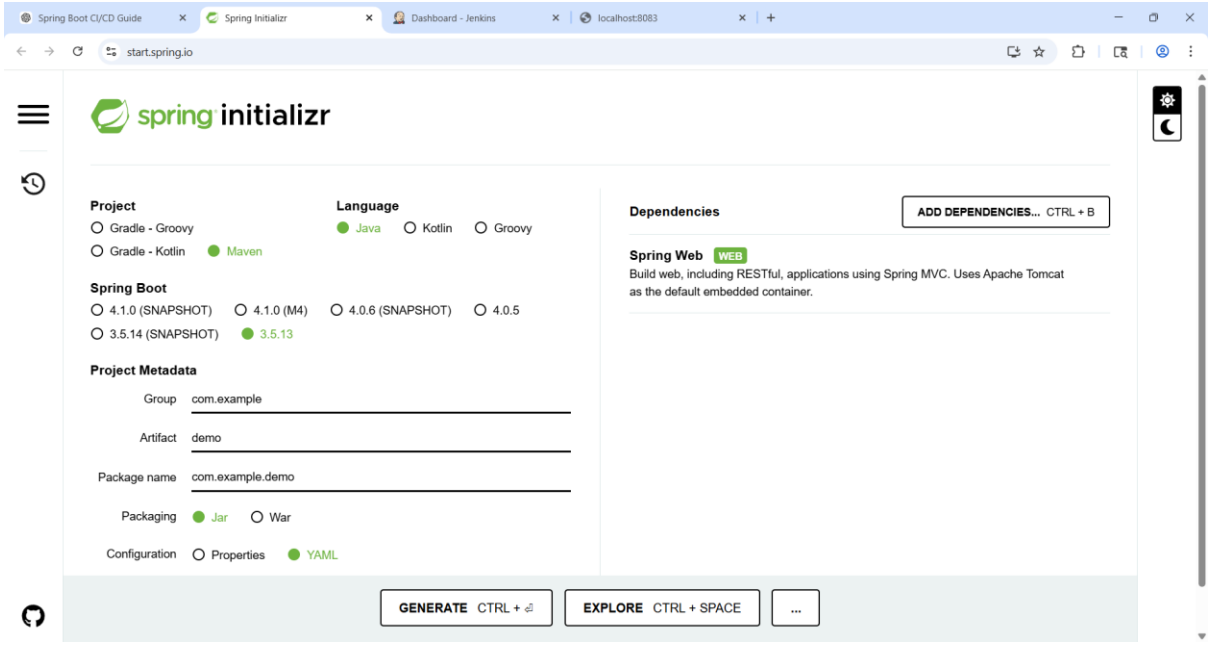
C: > Users > PREETHIKSHABALA V > Downloads > demo > src > main > resources > a

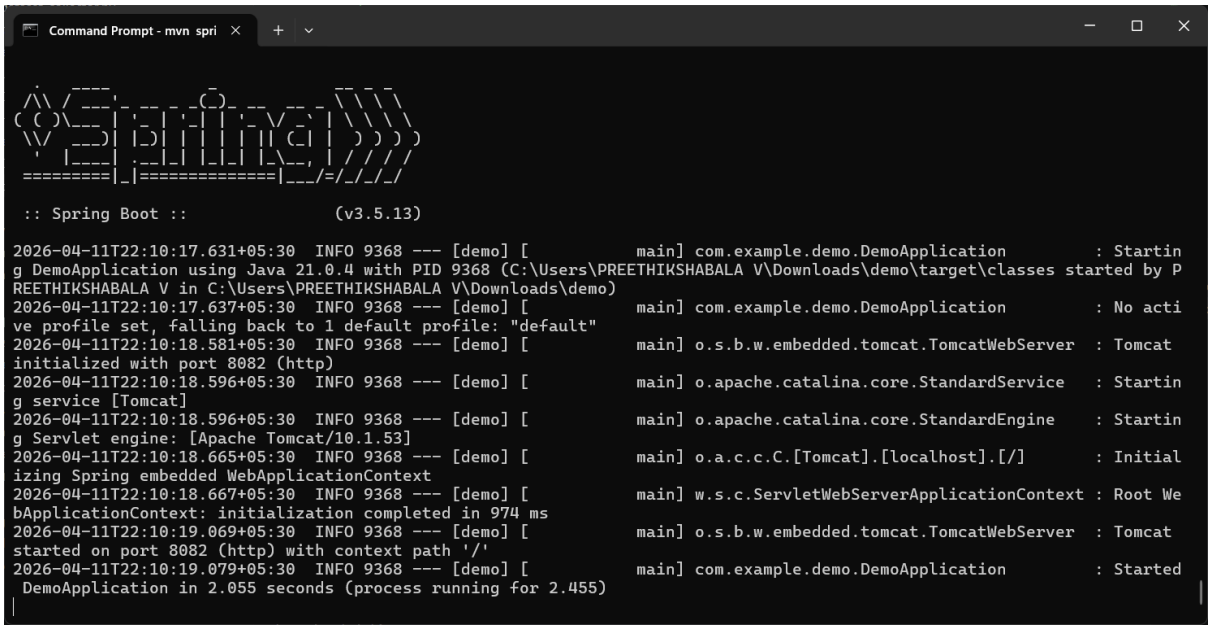
```yaml
spring:
  application:
    name: demo
server:
  port: 8082
```

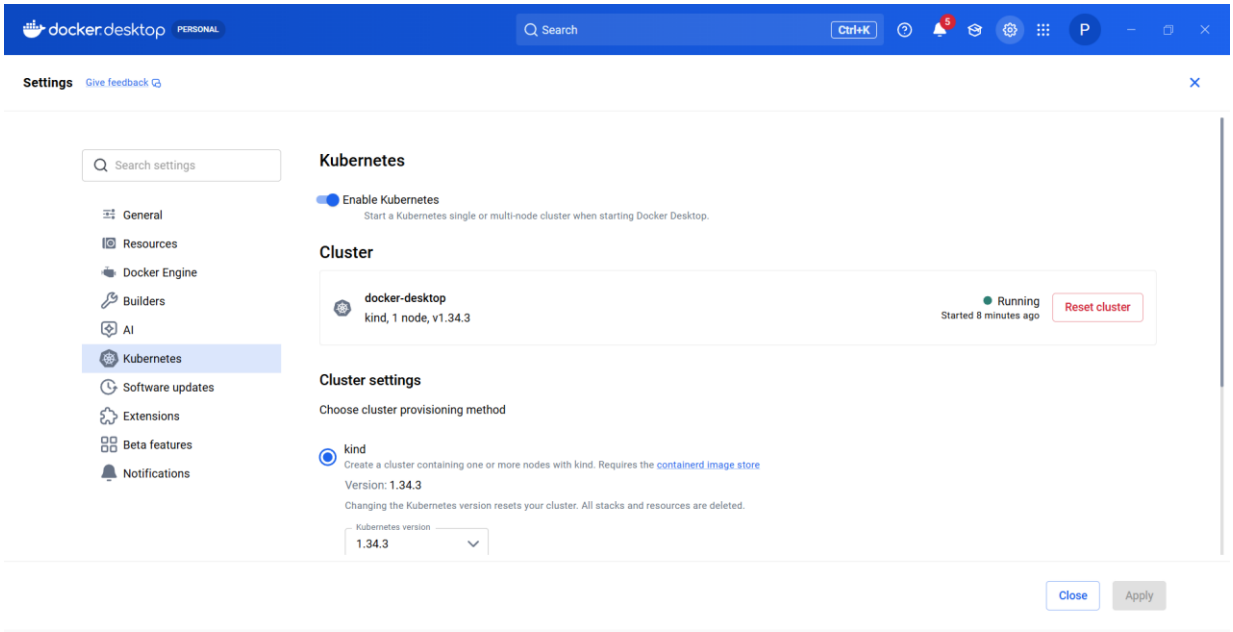
o (1).zip.369\... | HelloController.java | application.yaml | Dockerfile | deploy

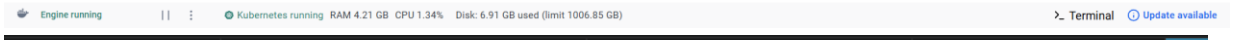
C: > Users > PREETHIKSHABALA V > Downloads > demo > Dockerfile

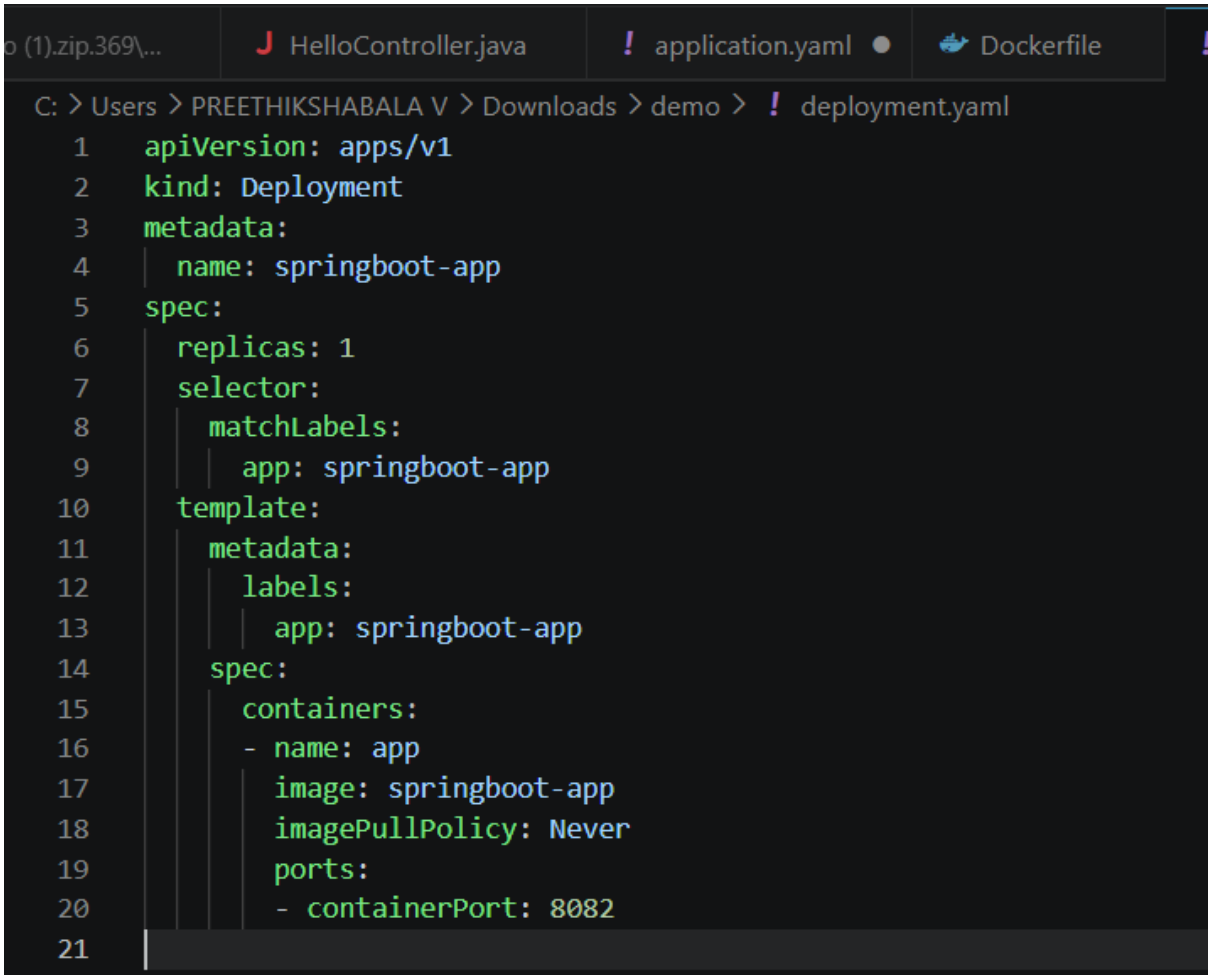
```dockerfile
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/*.jar app.jar
ENTRYPOINT ["java","-jar","app.jar"]
```


Spring Boot CI/CD Guide
Spring Initializr
Dashboard - Jenkins
localhost:8083
start.spring.io
spring initializr
Project
Gradle - Groovy
Gradle - Kotlin
Maven
Language
Java
Kotlin
Groovy
Spring Boot
4.1.0 (SNAPSHOT)
4.1.0 (M4)
4.0.6 (SNAPSHOT)
4.0.5
3.5.14 (SNAPSHOT)
3.5.13
Project Metadata
Group
com.example
Artifact
demo
Package name
com.example.demo
Packaging
Jar
War
Configuration
Properties
YAML
Dependencies
ADD DEPENDENCIES... CTRL + B
Spring Web
WEB
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.
GENERATE CTRL + ⏎
EXPLORE CTRL + SPACE
...


Command Prompt - mvn spri
:: Spring Boot :: (v3.5.13)
2026-04-11T22:10:17.631+05:30 INFO 9368 --- [demo] [main] com.example.demo.DemoApplication : Starting DemoApplication using Java 21.0.4 with PID 9368 (C:\Users\PREETHIKSHABALA V\Downloads\demo\target\classes started by PREETHIKSHABALA V in C:\Users\PREETHIKSHABALA V\Downloads\demo)
2026-04-11T22:10:17.637+05:30 INFO 9368 --- [demo] [main] com.example.demo.DemoApplication : No active profile set, falling back to 1 default profile: "default"
2026-04-11T22:10:18.581+05:30 INFO 9368 --- [demo] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8082 (http)
2026-04-11T22:10:18.596+05:30 INFO 9368 --- [demo] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-04-11T22:10:18.596+05:30 INFO 9368 --- [demo] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.53]
2026-04-11T22:10:18.665+05:30 INFO 9368 --- [demo] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-04-11T22:10:18.667+05:30 INFO 9368 --- [demo] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 974 ms
2026-04-11T22:10:19.069+05:30 INFO 9368 --- [demo] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8082 (http) with context path '/'
2026-04-11T22:10:19.079+05:30 INFO 9368 --- [demo] [main] com.example.demo.DemoApplication : Started DemoApplication in 2.055 seconds (process running for 2.455)








```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: springboot-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: springboot-app
  template:
    metadata:
      labels:
        app: springboot-app
    spec:
      containers:
      - name: app
        image: springboot-app
        imagePullPolicy: Never
        ports:
        - containerPort: 8082
```

```
PS C:\Users\PREETHIKSHABALA V\Downloads\demo> docker build -t springboot-app .
 => => sha256:a1e6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45 22.97MB / 22.97MB
 => => sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904 2.28kB / 2.28kB
 => => sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6 29.73MB / 29.73MB
 => => extracting sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6
 => => extracting sha256:a1e6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45
 => => extracting sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4
 => => extracting sha256:a149c623348dfa371df99c86694ffdf860874791706585045a923ea4e6224b0d
 => => extracting sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904
 => [internal] load build context
 => => transferring context: 21.09MB
 => [2/3] WORKDIR /app
 => [3/3] COPY target/*.jar app.jar
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:583329698c84846740f4c441fb1dddfe1df7d93b58fd67d83e4790625d3d1ca9
 => => exporting config sha256:b4a1d74a12c429b279d2c29d4f811cb8d4399c16dde9ddadf2c265d63c8f0002
 => => exporting attestation manifest sha256:6bde6f11e6f1c5a6276d30ee58eb0ba7595342df64058d90f3f12b498536bb
 => => exporting manifest list sha256:4bc56237774a5a20bc90817de6b8239b55749c837b92a5256a5575437a603452
 => => naming to docker.io/library/springboot-app:latest
 => => unpacking to docker.io/library/springboot-app:latest
```

```
PS C:\Users\PREETHIKSHABALA V\Downloads\demo> docker run -p 8083:8082 springboot-app

  .   ____          _            __ _ _
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 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v3.5.13)

2026-04-11T16:56:11.391Z  INFO 1 --- [demo] [           main] com.example.demo.DemoApplication         : Starting Dem
oApplication v0.0.1-SNAPSHOT using Java 17.0.18 with PID 1 (/app/app.jar started by root in /app)
2026-04-11T16:56:11.397Z  INFO 1 --- [demo] [           main] com.example.demo.DemoApplication         : No active pr
ofile set, falling back to 1 default profile: "default"
2026-04-11T16:56:12.735Z  INFO 1 --- [demo] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initi
alized with port 8082 (http)
2026-04-11T16:56:12.758Z  INFO 1 --- [demo] [           main] o.apache.catalina.core.StandardService   : Starting ser
vice [Tomcat]
2026-04-11T16:56:12.759Z  INFO 1 --- [demo] [           main] o.apache.catalina.core.StandardEngine    : Starting Ser
vlet engine: [Apache Tomcat/10.1.53]
2026-04-11T16:56:12.812Z  INFO 1 --- [demo] [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing
 Spring embedded WebApplicationContext
2026-04-11T16:56:12.814Z  INFO 1 --- [demo] [           main] w.s.c.ServletWebServerApplicationContext : Root WebAppl
icationContext: initialization completed in 1314 ms
2026-04-11T16:56:13.376Z  INFO 1 --- [demo] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat start
ed on port 8082 (http) with context path '/'
2026-04-11T16:56:13.414Z  INFO 1 --- [demo] [           main] com.example.demo.DemoApplication         : Started Demo
Application in 2.87 seconds (process running for 3.962)
2026-04-11T16:58:52.884Z  INFO 1 --- [demo] [nio-8082-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing
 Spring DispatcherServlet 'dispatcherServlet'
2026-04-11T16:58:52.885Z  INFO 1 --- [demo] [nio-8082-exec-1] o.s.web.servlet.DispatcherServlet        : Initializing
 Servlet 'dispatcherServlet'
2026-04-11T16:58:52.889Z  INFO 1 --- [demo] [nio-8082-exec-1] o.s.web.servlet.DispatcherServlet        : Completed in
itialization in 3 ms
```

localhost:8083

App Running Successfully 🚀

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl get nodes
NAME                     STATUS   ROLES           AGE     VERSION
desktop-control-plane    Ready    control-plane   6m14s   v1.34.3

C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl apply -f deployment.yaml
deployment.apps/springboot-app created
```

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl get pods
NAME                              READY   STATUS              RESTARTS   AGE
springboot-app-68645dfd47-9zk5c   0/1     ErrImageNeverPull   0          2m49s
```

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl expose deployment springboot-app --type=NodePort --port=8082
service/springboot-app exposed
```

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl get svc
NAME             TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
kubernetes       ClusterIP   10.96.0.1      <none>        443/TCP          10m
springboot-app   NodePort    10.96.68.191   <none>        8082:31026/TCP   19s
```

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>kubectl get pods
NAME                              READY   STATUS              RESTARTS   AGE
springboot-app-68645dfd47-9zk5c   0/1     ErrImageNeverPull   0          6m33s
```

```
C:\Users\PREETHIKSHABALA V\Downloads\demo>docker images
                                                                                                  i Info →   U  In Use
IMAGE                                          ID            DISK USAGE   CONTENT SIZE   EXTRA
docker/desktop-cloud-provider-kind:v0.5.0      4ad59ce20658       595MB          162MB    U
docker/desktop-containerd-registry-mirror:v0.0.3  179a9ea3ae47      50.3MB         16.1MB    U
envoyproxy/envoy:v1.36.4                       08acc53db6f4       248MB         62.1MB
hello-world:latest                             452a468a4bf9      25.9kB         9.49kB
kindest/node:v1.34.3                           08497ee19eac      1.35GB          395MB    U
lms-app:v1                                     a7eaf5745587      1.58GB          397MB    U
lms-app:v2                                     47f84aaa8a8c       1.6GB          406MB    U
lms-app:v3                                     eaf9b4f07b88       1.6GB          406MB    U
preethiksha/lms-app:v1                         a7eaf5745587      1.58GB          397MB    U
springboot-app:latest                          4bc56237774a       676MB          217MB    U
```